

The 2D Stochastic Heat Equation in the strong disorder regime

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Rome Workshop on Log-Correlated fields and SPDEs

Villa Mondragone (Roma Tor Vergata) • 21 September 2026

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Outline

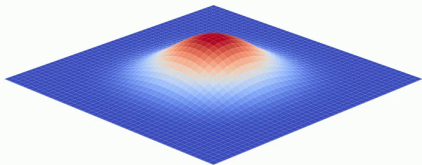
1. Stochastic Heat Equation
2. Critical regime
3. Strong disorder
4. Directed polymers
5. Sketch of the proof
6. Conclusions

Heat Equation

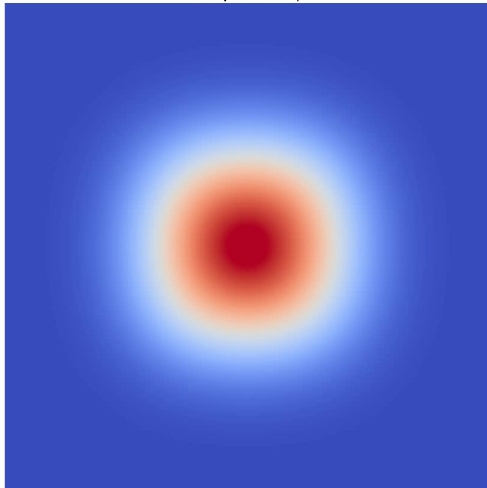
$$\partial_t u(t, x) = \Delta_x u(t, x)$$

(video by Q. Berger)

Density after n=14448 steps



Heat map after n=14448 step



In a nutshell

Stochastic Heat Equation

ill-defined in space dim. $d \geq 2$

$$\partial_t u(t, x) = \underbrace{\Delta_x u(t, x)}_{\sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} u(t, x)} + \underbrace{\beta \xi(t, x)}_{\text{"space-time white noise"} \atop \text{Gaussian cov. } \delta(t - t') \delta(x - x')} \cdot u(t, x)$$

There is a **non-trivial solution** in critical space dimension $d = 2$

Critical 2D Stochastic Heat Flow (SHF)

We study its properties for **strong disorder** β

Singular SPDEs



Statistical Mechanics

The role of dimension

$$\partial_t u(t, x) = \Delta_x u(t, x) + \beta \xi(t, x) u(t, x)$$

Space-time blow-up $\tilde{u}(t, x) := u(\delta^2 t, \delta x)$

$$\partial_t \tilde{u}(t, x) = \Delta_x \tilde{u}(t, x) + \delta^{\frac{2-d}{2}} \beta \tilde{\xi}(t, x) \tilde{u}(t, x)$$

As $\delta \downarrow 0$ the noise strength $\begin{cases} \text{vanishes} & d < 2 & \text{sub-critical} \\ \text{unchanged} & d = 2 & \text{critical} \\ \text{diverges} & d > 2 & \text{super-critical} \end{cases}$

We will henceforth focus on **critical dimension** $d = 2$

Regularization

Regularized noise $\xi_1(t, x)$ $\rightsquigarrow$ well-defined solution $u_1(t, x)$

(say: discretized on scale 1)

$$\partial_t u_1(t, x) = \Delta_x u_1(t, x) + \beta \xi_1(t, x) u_1(t, x) \quad u_1(0, x) \equiv 1 \text{ (say)}$$

Diffusive rescaling $u_N(t, x) := u_1(Nt, \sqrt{N}x)$ (large-scale of u_1)

$$\partial_t u_N(t, x) = \Delta_x u_N(t, x) + \beta \xi_N(t, x) u_N(t, x) \quad u_N(0, x) \equiv 1$$

$$\xi_N(t, x) := N \xi_1(Nt, \sqrt{N}x) \xrightarrow[N \rightarrow \infty]{d} \text{white noise } \xi(t, x)$$

Convergence?

Two interpretations for $u_N(t, x)$

- ▶ Large-scale behavior of **regularized SHE solution** $u_1(t, x)$
- ▶ Approximation of **ill-defined original SHE**

Convergence of $u_N(t, \varphi) = \int_{\mathbb{R}^2} u_N(t, x) \varphi(x) dx$? $[\varphi \in C_c: \varphi \geq 0, \int \varphi = 1]$

NO! Theorem

[Berger C. Turchi 25]

For every fixed $\beta > 0$ and $t > 0$: $u_N^\beta(t, \varphi) \xrightarrow[N \rightarrow \infty]{d} 0$ (finer results later)

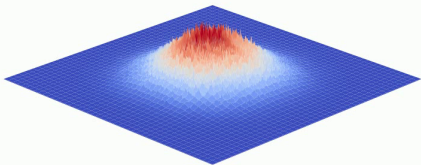
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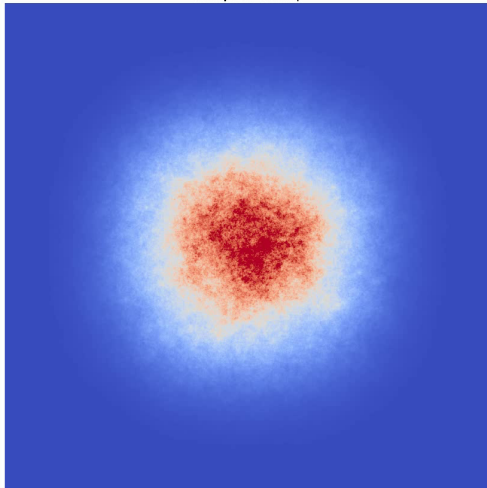
Discretization: β very small

(by Q. Berger)

Density after $n=14448$ steps

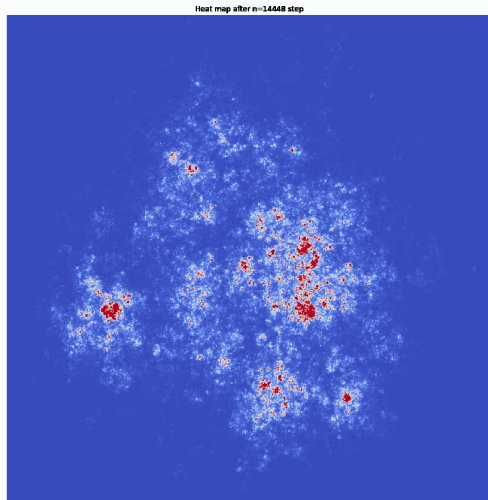
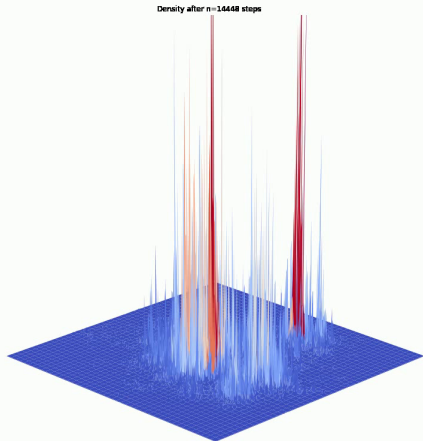


Heat map after $n=14448$ step



Discretization: β “critical”

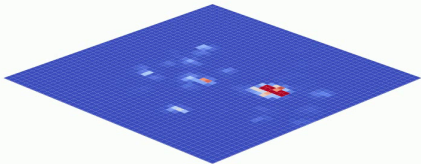
(by Q. Berger)



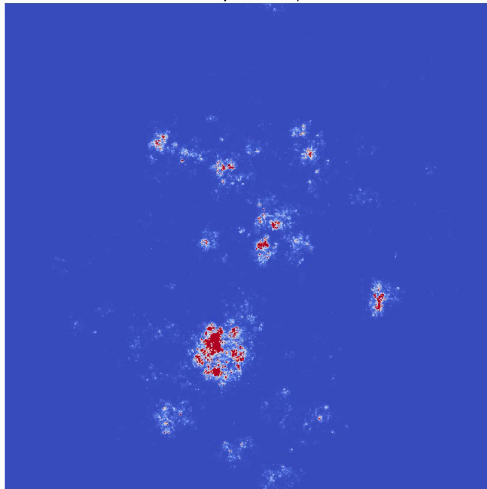
Discretization: β “supercritical”

(by Q. Berger)

Density after $n=14448$ steps



Heat map after $n=14448$ step



Renormalization

Why $u_N^\beta(t, \varphi) \rightarrow 0$ for fixed $\beta > 0$?

- ▶ Mean is constant $\mathbb{E}[u_N^\beta(t, \varphi)] = \int \varphi \equiv 1$ [martingale]
- ▶ 2nd moment explodes $\mathbb{E}[u_N^\beta(t, \varphi)^2] \rightarrow \infty \quad \forall \text{ fixed } \beta > 0$

Tame moment explosion: critical tuning $\beta^2 = \beta_N^2 \sim \frac{\pi}{\log N} \rightarrow 0$

- ▶ $\mathbb{E}[u_N^{\beta_N}(t, \varphi)^2] \rightarrow K_t(\varphi, \varphi) > 0$ [Bertini Cancrini 98] [C.S.Z. 19]
- ▶ Higher moments converge too [C.S.Z. 19] [Gu Quastel Tsai 21]

Convergence of $u_N^{\beta_N}(t, \varphi)$?

The critical 2D Stochastic Heat Flow

Theorem

[C.S.Z. 23]

Critical window $\beta_N^2 = \frac{\pi}{\log N} \left(1 + \frac{\vartheta}{\log N} \right)$ for some $\vartheta \in \mathbb{R}$

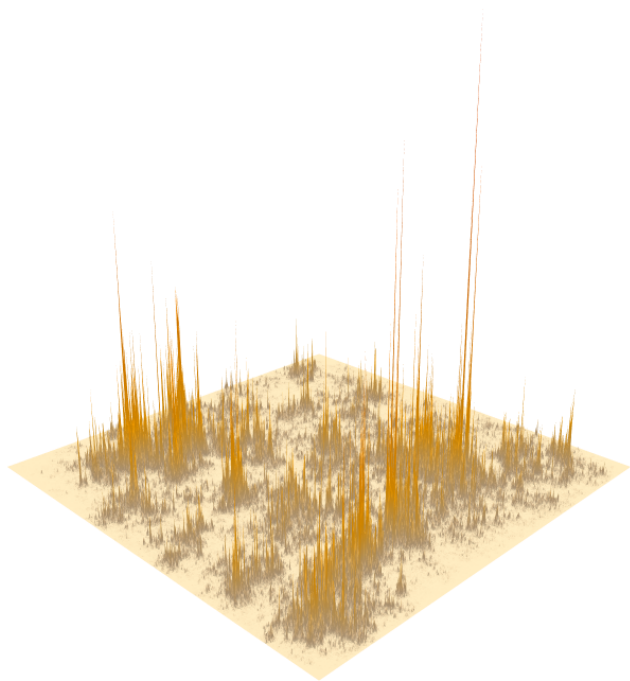
Convergence of u_N to a **unique** and **non-trivial limit** SHF^ϑ

$$\left(u_N^{\beta_N}(t, x) dx \right)_{t \geq 0} \xrightarrow[N \rightarrow \infty]{d} \left(\text{SHF}^\vartheta(t, dx) \right)_{t \geq 0}$$

$\text{SHF}^\vartheta =$ critical 2D **Stochastic Heat Flow (SHF)** = process of **random measures** on $\mathbb{R}^2$

Basic Properties of SHF

- ▶ $\text{SHF}^{\vartheta}(t, dx) \neq \text{Gaussian}$ (positive random measure)
- ▶ $\text{SHF}^{\vartheta}(t, dx) \neq \text{GMC} \simeq \exp(\text{Gaussian})$ on $\mathbb{R}^2$ [CSZ 23]
Conditional GMC structure on path space [Clark–Tsai 25]
- ▶ Moment formulas [CSZ 19] [Gu–Quastel–Tsai 21] [Ganguly–Nam 25]
 $\rightsquigarrow$ [Cosco, Nakajima, Zeitouni] [Lygkonis, Zygouras] [Liu, Zygouras] ...
- ▶ Axiomatic characterisation by independence and moments [Tsai 24]
- ▶ Diffusive rescaling $\text{SHF}^{\vartheta}(\alpha t, d(\sqrt{\alpha}x)) \stackrel{d}{=} \alpha \text{SHF}^{\vartheta+\log \alpha}(t, dx)$



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Supercritical regime

Regularized Stochastic Heat Equation

$$\partial_t u_N(t, x) = \Delta u_N(t, x) + \beta_N \xi_N(t, x) u_N(t, x) \quad u_N(0, \cdot) \equiv 1$$

$$\xi_N \rightarrow \xi \text{ white noise} \quad u_N \rightarrow \text{SHF}^{\vartheta} \quad \text{critical } \beta = \beta_N \rightarrow 0$$

$$\beta^2 = \frac{\pi}{\log N} \left(1 - \frac{\vartheta}{\log N} \right)^{-1} = \frac{\pi}{\log N - \vartheta}$$

Strong disorder regime

(supercritical)

Any regime of $\vartheta \rightarrow \infty$

$$1 \ll \vartheta \leq \log N - O(1)$$

Main result: local extinction

Theorem

[Berger C. Turchi 25]

$$u_N(t, \varphi) \xrightarrow[N \rightarrow \infty]{d} 0 \quad \text{as soon as } \vartheta \rightarrow \infty \quad \text{SHF}^\vartheta(t, \varphi) \xrightarrow[\vartheta \rightarrow \infty]{d} 0$$

(incl. fixed $\beta > 0$)

We next give quantitative bounds via

Fractional moments

Truncated mean

$$\mathbb{E}[u_N(t, \varphi)^\gamma] \quad \text{with } \gamma \in (0, 1)$$

$$\mathbb{E}[u_N(t, \varphi) \wedge 1] = \mathbb{P}(u_N(t, \varphi) > U(0, 1))$$

We focus for simplicity on $t = 1$ and

$$\varphi(\cdot) = B_R(\cdot) = \frac{1}{\pi R^2} \mathbb{1}_{|\cdot| \leq R}$$

Quantitative bounds

Theorem

[Berger C. Turchi 25]

Fix any $R > 0$. There are c, C such that

$$c \exp(-C e^{\vartheta}) \leq \mathbb{E}[u_N(1, B_R) \wedge 1] \leq C \exp(-c e^{\vartheta})$$

Note that

$$e^{\vartheta} = e^{-\pi/\beta^2} N = O(1) \text{ for } \beta = \beta_N \text{ critical}$$

$\forall \varepsilon \in (0, 1)$ there are $c_\varepsilon, C_\varepsilon$ such that

$$c_\varepsilon \exp(-C e^{\vartheta}) \leq \mathbb{P}(u_N(1, B_R) \geq \varepsilon) \leq C_\varepsilon \exp(-c e^{\vartheta})$$

LB: 2nd moment method

UB: coarse-graining + change of measure

Spatial scale of mass escape

For fixed $c > 0$ define

$$R_{\vartheta}^c := \exp(c e^{\vartheta}) = \exp(c e^{-\pi/\beta^2} N) \quad = O(1) \text{ for } \beta = \beta_N \text{ critical}$$

Theorem

[Berger C. Turchi 25]

There are $0 < c_- \leq c_+ < \infty$ such that

$$u_N(1, R_{\vartheta}^c) \xrightarrow[N \rightarrow \infty]{d} \begin{cases} 0 & \text{if } c < c_- \\ 1 & \text{if } c > c_+ \end{cases}$$

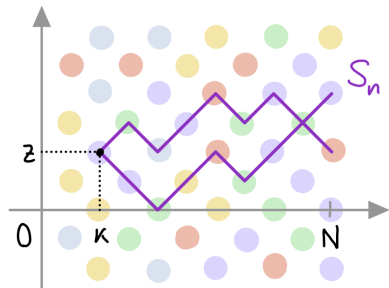
Conjecture: sharp transition $c_- = c_+$

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Directed polymers partition functions

- ▶ $S = (S_n)_{n \geq 0}$ simple random walk on $\mathbb{Z}^d$
- ▶ Independent Gaussians $\omega(n, x) \sim \mathcal{N}(0, 1)$
- ▶ $H(S, \omega) := \sum_{n=k+1}^N \omega(n, S_n)$



Partition Functions

$(k \in \mathbb{N}, z \in \mathbb{Z}^d)$

$$Z_{N,\beta}^\omega(k, z) = \mathbb{E} \left[e^{\beta H(S, \omega) - \frac{1}{2} \beta^2 (N-k)} \mid S_k = z \right]$$

$$\mathbb{E} [Z_{N,\beta}^\omega] = 1$$

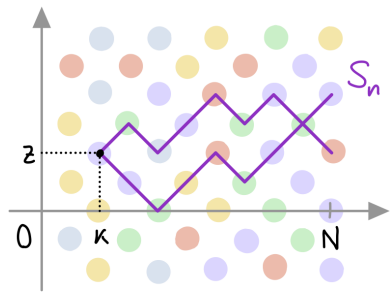
Partition functions and SHE

Partition functions $Z_{N,\beta}^\omega(k,z)$ are solutions of
discretised Stochastic Heat Equation

Diffusive rescaling (+ time reversal)

$$u_N(t,x) := Z_{N,\beta}^\omega(N(1-t), \sqrt{N}x)$$

$$\begin{cases} \partial_t u_N(t,x) = \Delta_x u_N(t,x) + \beta \underbrace{\xi_N(t,x)}_{\text{white noise discr. on scale } 1/\sqrt{N}} u_N(t,x) \\ u_N(0,x) \equiv 1 \end{cases} \quad (\text{disc-SHE})$$



Main results: local extinction and free energy

Theorem

[Berger C. Turchi 25]

$$c \exp(-C e^{\vartheta}) \leq \mathbb{E}[Z_{N,\beta}^{\omega}(B_{R\sqrt{N}}) \wedge 1] \leq C \exp(-c e^{\vartheta}) = C \exp(-c e^{-\pi/\beta^2} N)$$

Free energy (point-to-plane)

[Berger Lacoïn 17]

$$F(\beta) := \lim_{N \rightarrow \infty} \frac{1}{N} \log Z_{N,\beta}(0) = \exp\left(-\frac{\pi + o(1)}{\beta^2}\right) \quad \text{as } \beta \downarrow 0$$

Theorem

[Berger C. Turchi 25] [Berger Nakashima 26]

$$-C \exp\left(-\frac{\pi}{\beta^2}\right) \leq F(\beta) \leq -c \exp\left(-\frac{\pi}{\beta^2}\right)$$

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Proof of the UB: coarse-graining

$$\mathbb{E}[u_N^\beta(1, B_1) \wedge 1] = \mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \leq C \exp(-c e^\vartheta)$$

► Coarse-graining argument: $\begin{cases} \text{reduce } \vartheta \rightarrow \infty \text{ to fixed (large) } \vartheta > 0 \\ \text{replace } \exp(-c e^\vartheta) \text{ by any } f(\vartheta) \rightarrow 0 \end{cases}$

Key bound

$$\mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \leq \frac{C}{\vartheta} \quad \text{for fixed } \vartheta \in \mathbb{R}$$

Proof of the UB: change of measure

Change of scale: $\mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \leq \frac{C}{\epsilon} \mathbb{E}[Z_{N,\beta}(B_{\sqrt{\epsilon N}}) \wedge 1] \quad \forall \epsilon \in (0,1)$

Size-biased law $\tilde{\mathbb{P}}(d\omega) := Z(\omega) \mathbb{P}(d\omega) \quad Z(\omega) := Z_{N,\beta}(B_{\sqrt{\epsilon N}})$

Change of measure

cf. [Junk-Lacoin]

$$\mathbb{E}[Z \wedge 1] \leq \mathbb{P}(A) + \tilde{\mathbb{P}}(A^c) \quad \text{for any event } A$$

Goal: find some event A which is typical for $\mathbb{P}$ and atypical for $\tilde{\mathbb{P}}$

Optimal choice $A = \{Z > 1\}$ (but we don't know Z)

Proof of the UB: choice of a proxy

- ▶ Choose a “nice” r.v. X with $\mathbb{E}[X] = 0$ and $\tilde{\mathbb{E}}[X] \gg 1$ ($X \approx Z - 1$)
- ▶ Take $A = \{X > \frac{1}{2} \tilde{\mathbb{E}}[X]\} \rightsquigarrow \mathbb{P}(A) \leq 4 \frac{\text{Var}[X]}{\tilde{\mathbb{E}}[X]^2} \quad \tilde{\mathbb{P}}(A^c) \leq 4 \frac{\tilde{\text{Var}}[X]}{\tilde{\mathbb{E}}[X]^2}$
- ▶ Estimate $\text{Var}[X]$, $\tilde{\mathbb{E}}[X]$ (2nd moment) and $\tilde{\text{Var}}[X]$ (3rd moment)

We take X as a **manageable proxy** of $Z - 1$ restricting its **chaos expansion** to

$$I = \{(n_1, x_1), \dots, (n_k, x_k)\} \quad \text{with} \quad \begin{cases} \text{width}(I) = n_k - n_1 \leq \varepsilon N, \\ |I| = k \leq \log(\varepsilon N) \end{cases}$$

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Conclusions

Quantitative bounds for local extinction of SHF and directed polymers
strong disorder (or large time)

Equiv.: discretized SHE in the supercritical regime (including fixed $\beta > 0$)

Mass escape to infinity on the spatial scale $\exp(c e^{\vartheta}) = \exp(c N e^{-\pi/\beta^2})$

Robust proof based on coarse-graining + change of scale + change of measure

Grazie

Singularity and Regularity

- ▶ $\text{SHF}^{\vartheta}(t, dx)$ **singular** w.r.t. Lebesgue [CSZ 25+]
“not a function”
- ▶ $\text{SHF}^{\vartheta}(t, dx) \in \mathcal{C}^{-\kappa}$ for any $\kappa > 0$ $\rightsquigarrow$ non atomic
“barely not a function”
- ▶ **Small ball** $\text{SHF}^{\vartheta}(1, g_{\varepsilon^2}) = |\log \varepsilon|^{-1/2+o(1)}$ [Gu–Tsai 26]
+ Gaussian fluctuations
- ▶ **Strict positivity** $\text{SHF}^{\vartheta}(t, B) > 0 \quad \forall \text{ ball } B$ [Clark–Tsai 25]
[Nakashima 25]

SHF and noise

Regularized Stochastic Heat Equation

(discretized / mollified noise)

$$\partial_t u_N(t, x) = \Delta_x u_N(t, x) + \beta_N u_N(t, x) \xi_N(t, x)$$

$$\text{As } N \rightarrow \infty \quad u_N \longrightarrow \text{SHF}^\vartheta \quad \xi_N \longrightarrow \xi \quad \beta_N = \beta_N(\vartheta) \longrightarrow 0$$

Theorem

[C.–Donadini 25] [Gu–Tsai 25]

$$(u_N, \xi_N) \xrightarrow[\varepsilon \downarrow 0]{d} (\text{SHF}^\vartheta, \xi) \quad \text{with } \text{SHF}^\vartheta \text{ and } \xi \text{ independent}$$

$\rightsquigarrow$ **No SPDE** driven by ξ for SHF^ϑ

► Noise sensitivity [C.–Donadini 25]

► Black noise [Gu–Tsai 25]